

# Insider Threat Behavioral Intelligence System

## 1. Title: Insider Threat Behavioral Intelligence System

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## 2. Objective

Build an AI-powered Insider Threat Behavioral Intelligence System that continuously monitors employee activities, analyzes behavioral patterns, detects anomalies, evaluates insider risk levels, and generates security alerts for potential insider threats.

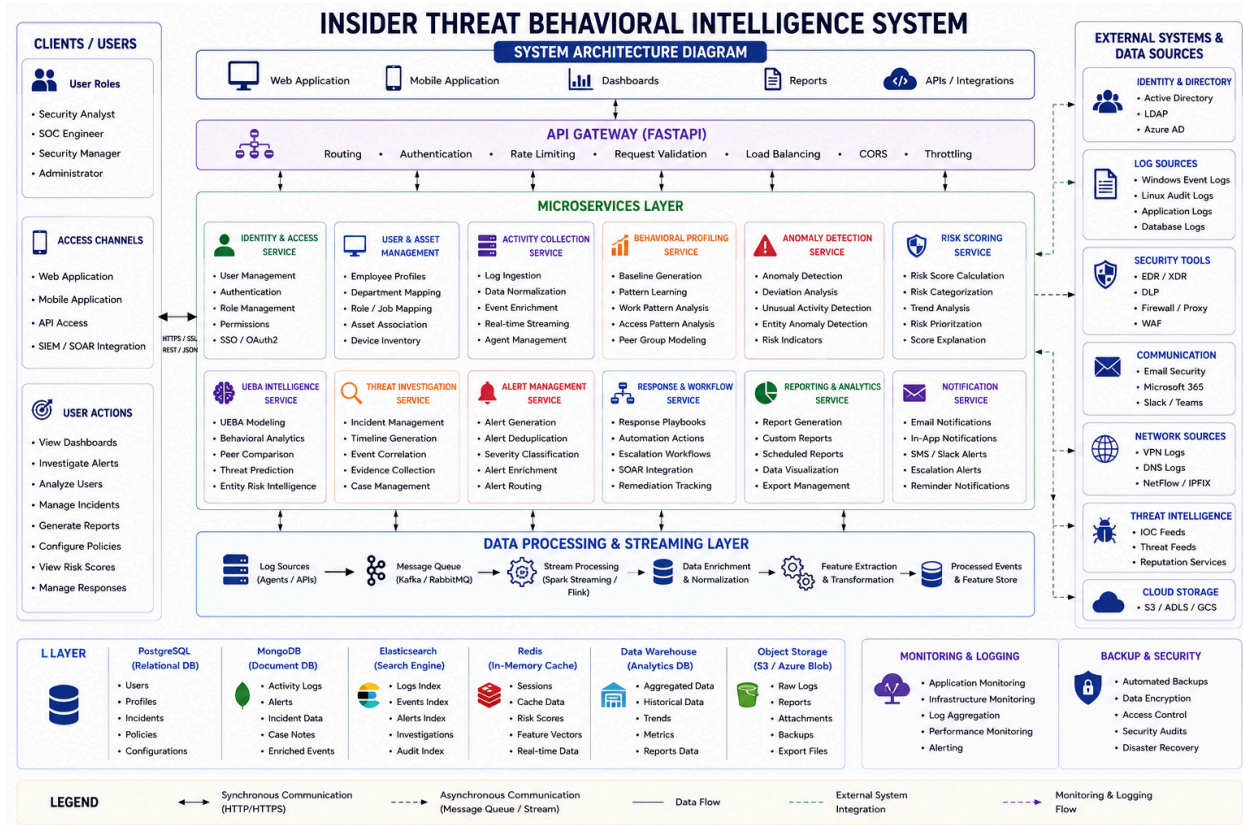
The platform helps organizations proactively identify suspicious behavior, prevent data breaches, detect account misuse, monitor privilege abuse, and strengthen enterprise security through behavioral analytics and AI-driven risk assessment.

The solution is designed for enterprises, financial institutions, healthcare organizations, government agencies, cybersecurity teams, and security operations centers (SOCs).

## Outcomes

- Developed and deployed an AI-powered insider threat detection platform.
  - Implemented secure authentication and role-based access control.
  - Built employee activity monitoring and behavioral profiling systems.
  - Developed anomaly detection and insider risk scoring engines.
  - Implemented user behavior analytics (UBA) and entity behavior analytics (UEBA).
  - Built threat investigation and incident management workflows.
  - Developed dashboards for security analysts and SOC teams.
  - Deployed the platform using Docker and cloud deployment platforms such as AWS or Azure.
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### 3. Architecture Diagram:



### 4. Modules to be Implemented

#### 1. User Authentication & Role-Based Access

- User registration and login
- JWT authentication
- OAuth2 login
- Role-based access control
- User profile management

#### Roles

- Security Analyst

- SOC Engineer
- Security Manager
- Administrator

## **2. Employee Identity & Profile Management**

- Employee onboarding
- User profile management
- Department mapping
- Role management
- Asset association

### **Employee Information**

- Employee ID
- Department
- Designation
- Manager
- Device Information
- Access Privileges

## **3. Activity Monitoring Engine**

- Login monitoring
- File access monitoring
- Application usage tracking
- Network activity monitoring
- Email activity tracking
- USB device monitoring

### **Monitored Activities**

- Login Events
- File Downloads
- File Uploads
- Data Transfers
- Email Activities
- Privilege Changes
- Remote Access Sessions

#### **4. Behavioral Profiling Engine**

- User behavior baseline generation
- Work pattern analysis
- Device usage analysis
- Access pattern analysis
- Productivity pattern monitoring

#### **Behavioral Indicators**

- Login Times
- Resource Access Frequency
- Device Usage
- Application Usage
- Data Transfer Volume
- Communication Patterns

#### **5. Anomaly Detection Engine**

- Behavioral anomaly detection
- Access anomaly detection
- Data exfiltration detection
- Privilege abuse detection

- Suspicious activity identification

## **Anomaly Categories**

- Unusual Login Time
- Abnormal Data Download
- Unauthorized Access Attempts
- Excessive File Transfers
- Suspicious Device Usage
- Insider Risk Indicators

## **6. Insider Risk Scoring Engine**

- Risk score generation
- Threat severity assessment
- Risk trend monitoring
- Threat prioritization
- Employee risk categorization

## **Risk Categories**

- Low Risk
- Medium Risk
- High Risk
- Critical Risk

## **Weighted Scoring Model**

Insider Risk Score =

- Behavioral Anomalies (35%)
- Privilege Misuse Indicators (25%)
- Data Access Violations (20%)

- Access Pattern Deviations (10%)
- Historical Security Events (10%)

## **7. Threat Investigation Module**

- Incident creation
- Threat timeline generation
- Activity correlation
- Investigation workflows
- Evidence management

### **Investigation Features**

- Activity Timeline
- User Risk History
- Device Analysis
- Event Correlation
- Threat Evidence Collection

## **8. UEBA Intelligence Engine**

- User behavior analytics
- Entity behavior analytics
- Peer group comparison
- Behavioral trend analysis
- Threat prediction

## **9. Alert & Incident Management System**

- Threat alerts
- Incident creation
- Escalation workflows

- Analyst assignment
- Resolution tracking

## **Alert Severity Levels**

- Informational
- Low
- Medium
- High
- Critical

## **10. Dashboard & Analytics**

### **Security Analyst Dashboard**

- Threat alerts
- Insider risk scores
- Investigation queue
- Incident summaries

### **SOC Dashboard**

- Security events
- Behavioral anomalies
- Active investigations
- Threat intelligence

### **Security Manager Dashboard**

- Organizational risk posture
- Risk trends
- Insider threat reports
- Compliance metrics

## **Admin Dashboard**

- User management
- Platform analytics
- System monitoring
- Audit reports

## **11. Notification & Escalation System**

- Insider threat alerts
- Investigation notifications
- Escalation alerts
- Compliance notifications
- Security event updates

## **12. Reports & Export System**

- Insider threat reports
- Behavioral analytics reports
- Investigation reports
- Compliance reports
- Risk assessment reports
- PDF export
- Excel export

## **13. Final Integration, Testing & Deployment**

- Frontend and backend integration
- API validation and testing
- End-to-end workflow testing
- Security testing
- Performance optimization



- Docker containerization
  - Production deployment
  - Monitoring and logging setup
  - Documentation and user guides
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## **5. Week-wise Module Implementation and High-Level Requirements**

### **Milestone 1: Week 1 & 2 — Project Initialization, Design Process & Core Setup**

#### **Tasks**

- Define project objectives and insider threat workflows.
- Design system architecture and database schema.
- Create UI wireframes and workflow planning.
- Setup frontend and backend environments.
- Implement authentication and role-based access system.
- Build employee profile management workflows.
- Configure activity log ingestion pipelines.

#### **Recommended Datasets**

##### **CERT Insider Threat Dataset**

Purpose:

- Insider threat detection
- User behavior analytics
- Security anomaly detection

##### **LANL Cyber Security Dataset**

Purpose:

- User activity monitoring
- Behavioral anomaly detection

## **CMU Insider Threat Dataset**

Purpose:

- Insider risk modeling
- Threat investigation workflows

## **Outcomes**

- Understand insider threat detection workflows.
- Learn behavioral analytics architecture.
- Build frontend and backend project initialization.
- Working authentication and activity monitoring system.

## **Milestone 2: Week 3 & 4 — Behavioral Analytics & Anomaly Detection**

### **Tasks**

- Implement behavioral profiling engine.
- Build anomaly detection workflows.
- Develop behavioral baselines.
- Create threat detection models.
- Generate anomaly reports.

### **Outcomes**

- Behavioral analytics engine operational.
- Anomaly detection workflows functional.
- Threat detection system completed.

## **Milestone 3: Week 5 & 6 — Risk Scoring & Threat Investigation**

## Tasks

- Implement insider risk scoring engine.
- Build UEBA intelligence workflows.
- Develop threat investigation modules.
- Generate risk analytics.
- Create security dashboards.

## Outcomes

- Insider risk scoring engine operational.
- Investigation workflows functional.
- Risk analytics dashboards completed.

## Milestone 4: Week 7 & 8 — Analytics, Testing & Deployment

### Tasks

- Build executive dashboards.
- Add reports and visualization modules.
- Implement testing and validations.
- Deploy platform using Docker and cloud services.
- Prepare final documentation and presentation.

### Outcomes

- Fully deployed production-ready platform.
  - Insider threat intelligence systems operational.
  - Complete end-to-end threat detection workflow demonstrable.
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## 6. Evaluation Criteria

### Milestone 1 (Week 2)

- Project initialization completed.
- Authentication implemented.
- Activity monitoring operational.
- Log ingestion workflows integrated.

### **Milestone 2 (Week 4)**

- Behavioral analytics engine operational.
- Anomaly detection workflows functional.
- Threat detection system implemented.

### **Milestone 3 (Week 6)**

- Risk scoring engine operational.
- Investigation workflows functional.
- Security analytics completed.

### **Milestone 4 (Week 8)**

- Fully deployed frontend and backend.
  - Dashboards and reporting systems operational.
  - End-to-end insider threat intelligence workflow demonstrated.
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## **7. Tools & Tech Stack**

### **Programming Language**

#### **Backend**

- Python
- FastAPI

#### **Frontend**

- JavaScript

- React.js

## **Database**

### **Primary Database**

- PostgreSQL

### **Secondary Database**

- MongoDB

## **AI & Machine Learning**

- Scikit-learn
- XGBoost
- Isolation Forest
- TensorFlow
- PyTorch
- Pandas
- NumPy

## **Behavioral Analytics & Security**

- UEBA Models
- Anomaly Detection Models
- Risk Scoring Models
- Graph Analytics

## **Security Log Sources**

- Active Directory Logs
- Windows Event Logs
- Linux Audit Logs
- VPN Logs

- Firewall Logs
- Email Security Logs
- Endpoint Security Logs

## **Search & Analytics**

- Elasticsearch
- Kibana
- OpenSearch

## **Cloud & DevOps**

- Docker
- AWS / Azure

## **Libraries & Frameworks**

- FastAPI
- React.js
- Next.js
- Tailwind CSS
- JWT Authentication
- Chart.js
- Plotly

## **Dev & Deployment Tools**

- VS Code
  - Git & GitHub
  - Docker & Docker Compose
  - GitHub Actions
  - Postman
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## **8. Performance Metrics**

### **Behavioral Analytics Metrics**

- Anomaly detection accuracy
- Behavioral baseline quality
- Risk classification accuracy

### **Insider Threat Metrics**

- Threat detection precision
- Threat detection recall
- False positive rate
- Threat investigation efficiency

### **Risk Scoring Metrics**

- Risk score consistency
- Risk categorization accuracy
- Threat prioritization effectiveness

### **Security Operations Metrics**

- Mean Time To Detect (MTTD)
- Mean Time To Investigate (MTTI)
- Mean Time To Respond (MTTR)

### **System Performance Metrics**

- API response time
  - Dashboard loading speed
  - Alert generation latency
  - Concurrent user handling capacity
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## **9. Example Quantitative Goals**

### **Insider Threat Detection**

Detect suspicious employee behavior before security incidents occur.

### **Behavioral Intelligence**

Build accurate behavioral baselines and identify deviations effectively.

### **Threat Investigation**

Accelerate security investigations through automated correlation and risk scoring.

### **Security Operations**

Improve SOC efficiency through intelligent alert prioritization.

### **Platform Performance**

Support large-scale enterprise activity monitoring and behavioral analytics while maintaining stable performance and responsiveness.